



Department of

Information and Communication Technology

M.Sc. (Engg.) in ICT (2024-2025)

ICT 6127: Advanced TCP/IP Inter-networking

A Brief Intro on

Transmission Control Protocol (TCP)

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OUTLINE

- ☐ TCP in OSI and Its use in online payment

- ☐ Multiplexing And Demultiplexing

- ☐ Addressing and Header Format

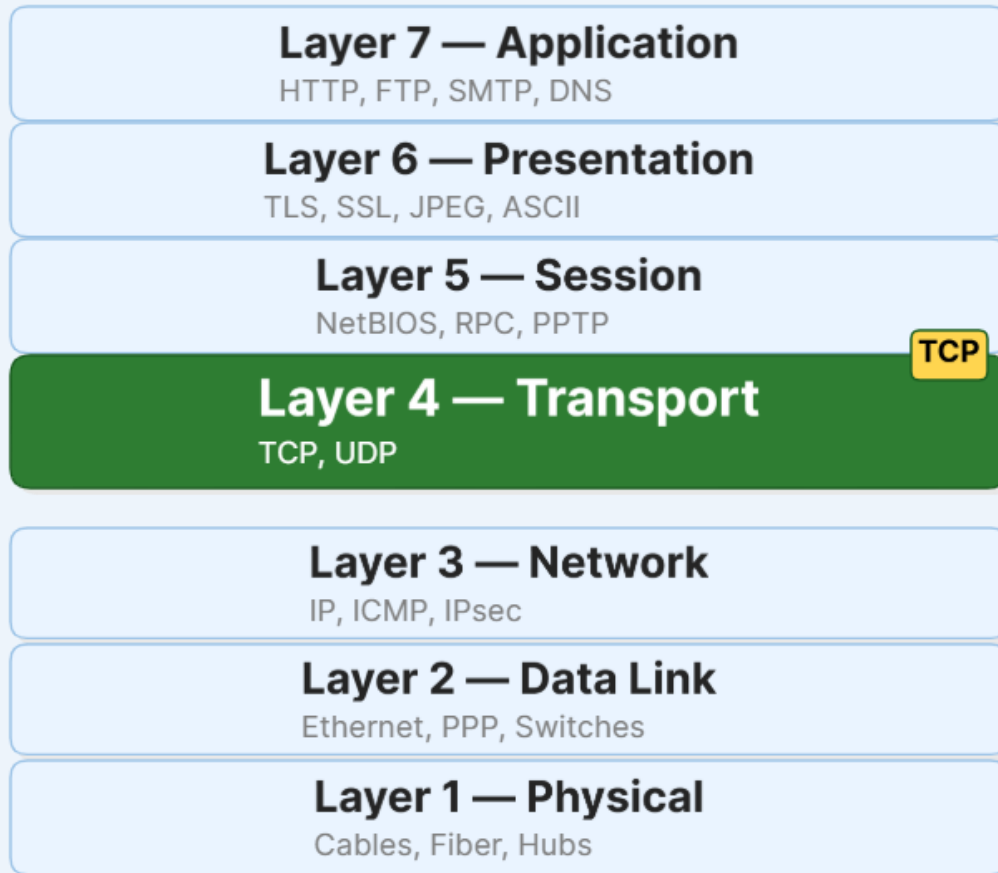
- ☐ Connection And Termination

- ☐ State Transition

- ☐ Application And Limitations

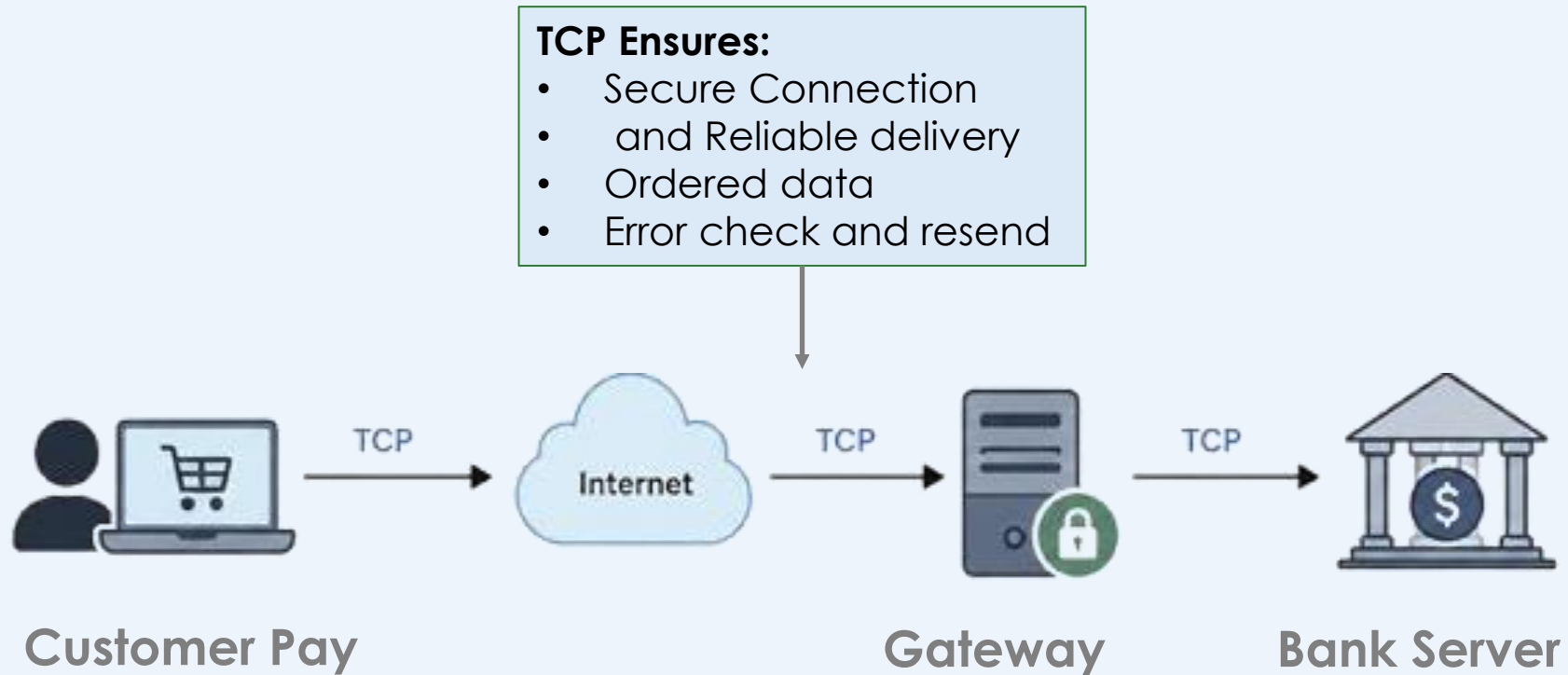
- ☐ References

TCP IN OSI

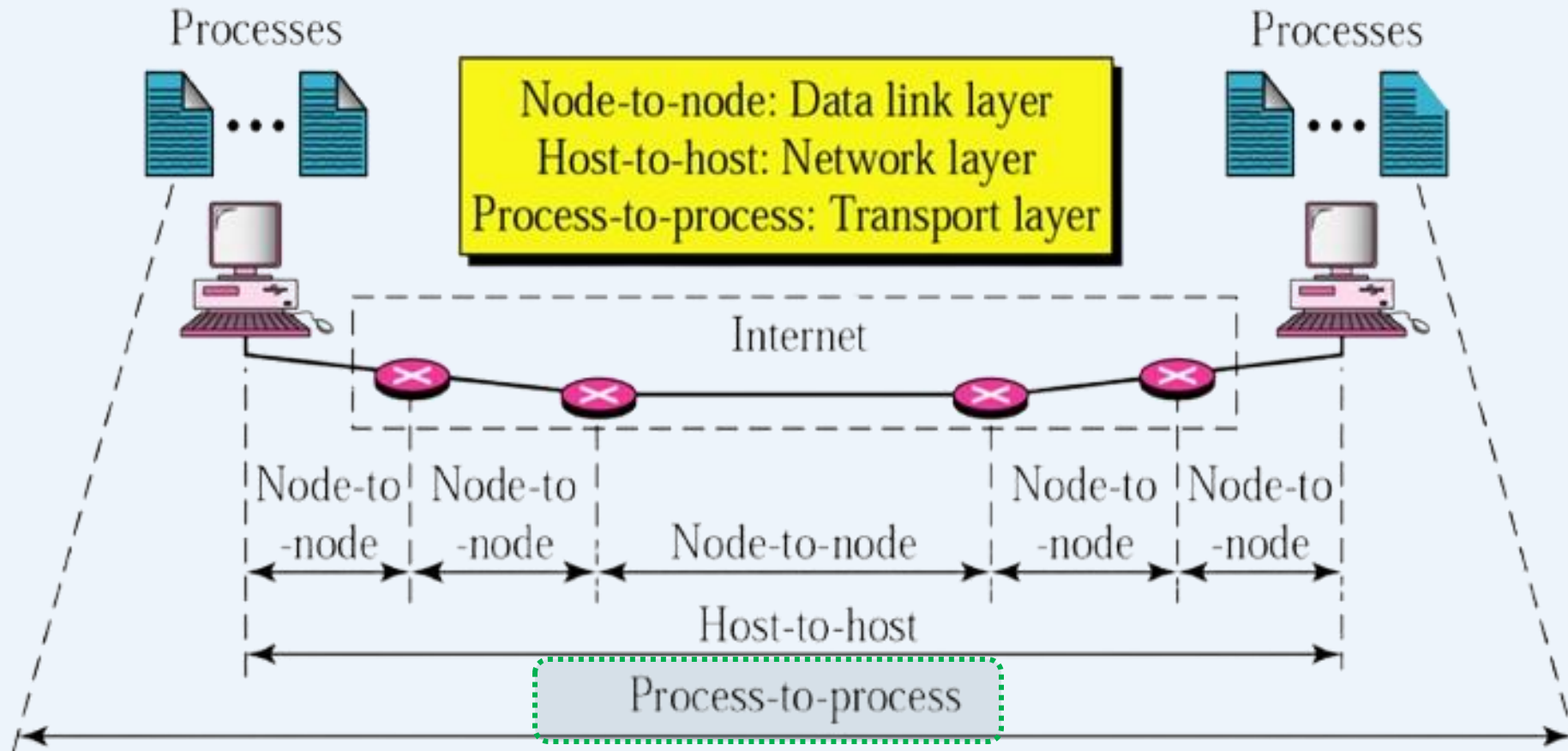


TCP: Transmission Control Protocol
- is a **connection-oriented transport layer protocol** that ensures reliable, ordered, and error-checked delivery of data between devices over a network.

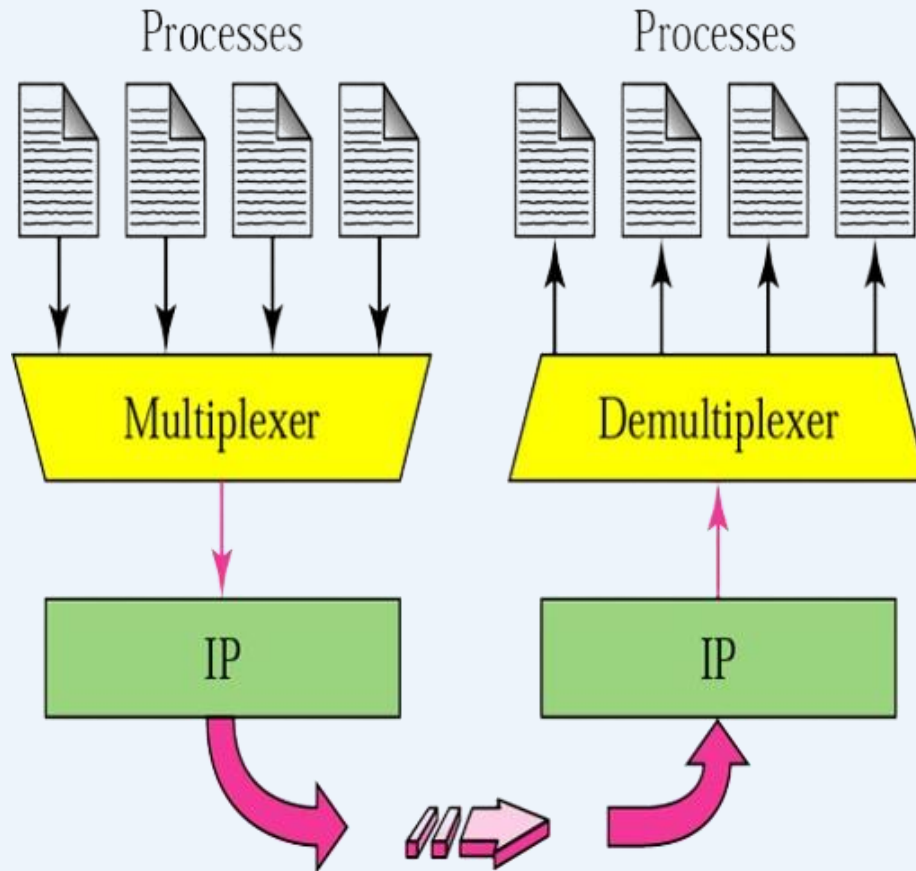
TCP IN BANKING



TRANSPORT LAYER



MULTIPLEXING AND DEMULTIPLEXING



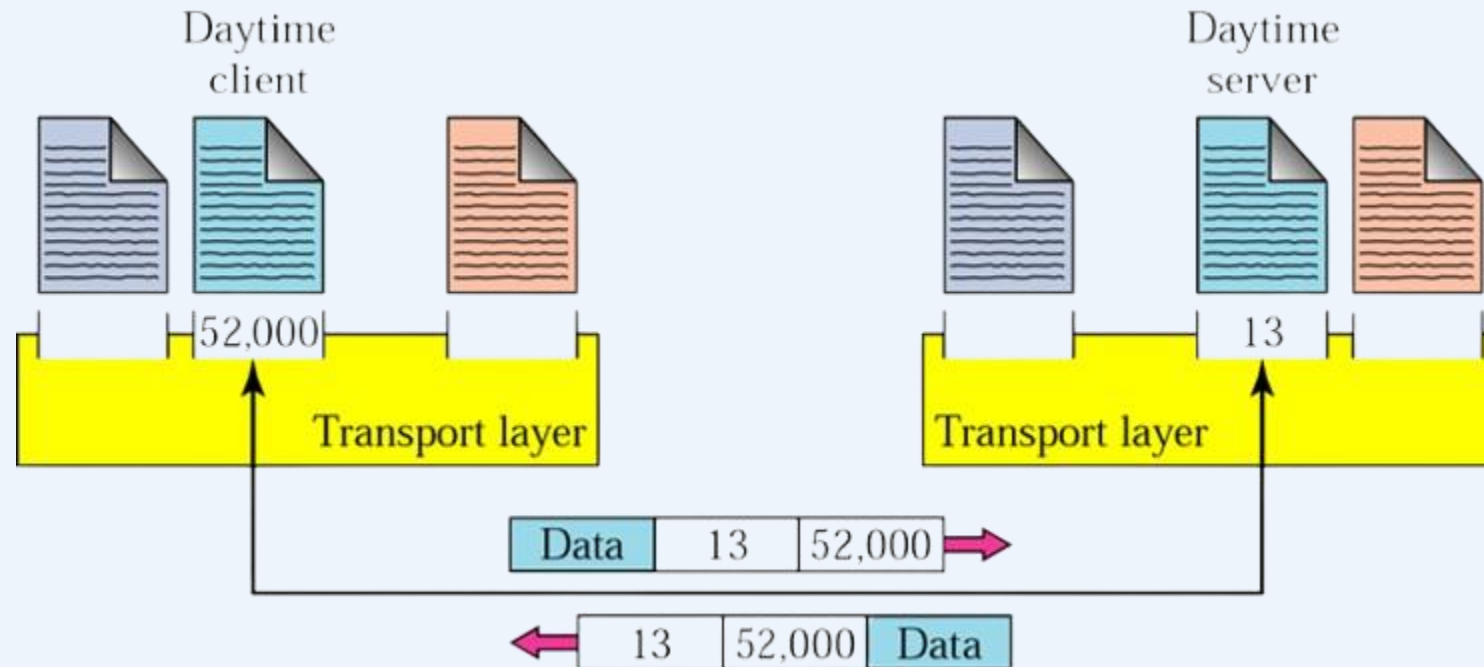
Multiplexing

- process of gathering data chunks from different processes at the sender
- **Processes** → **TCP Protocol** → **Stream**

Demultiplexing

- Error checking
- Header removal
- Delivery
- **Stream** → **TCP Headers** → **Processes**

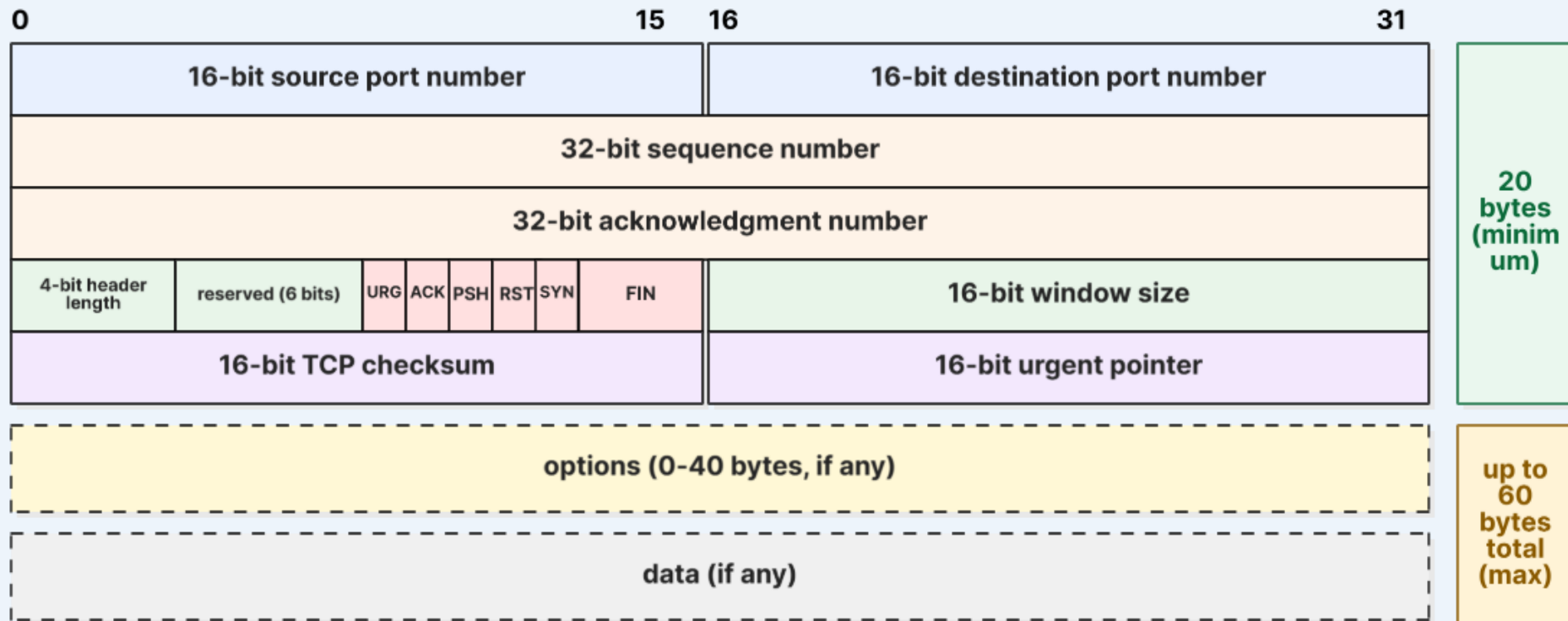
TCP ADDRESSING



Addresses

- Data link layer → MAC address
- Network layer → IP address
- **Transport layer → *Port number* (choose among multiple processes running on destination host)**

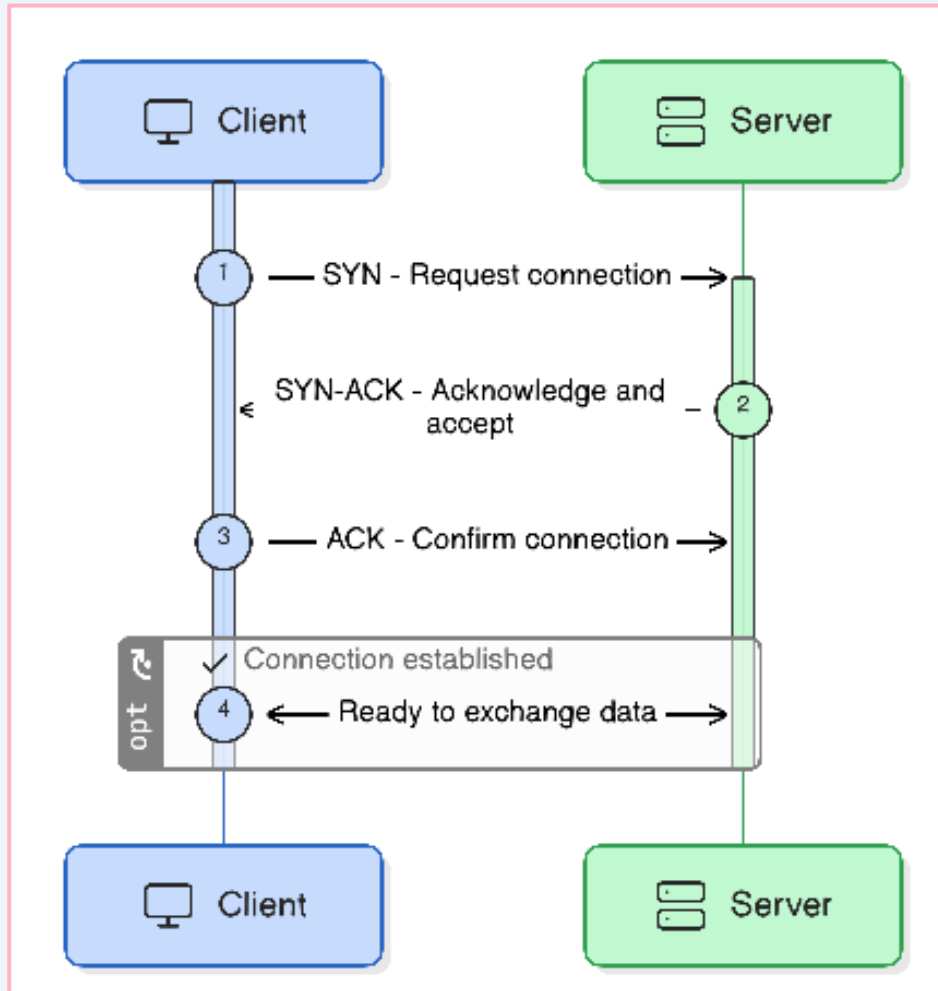
TCP HEADER FORMAT



- **Header length:** minimum **20 bytes** (no options) — **maximum 60 bytes** (with up to 40 bytes of options).
- The 4-bit *header length* field specifies the size in **32-bit words** (5–15).



TCP: CONNECTION



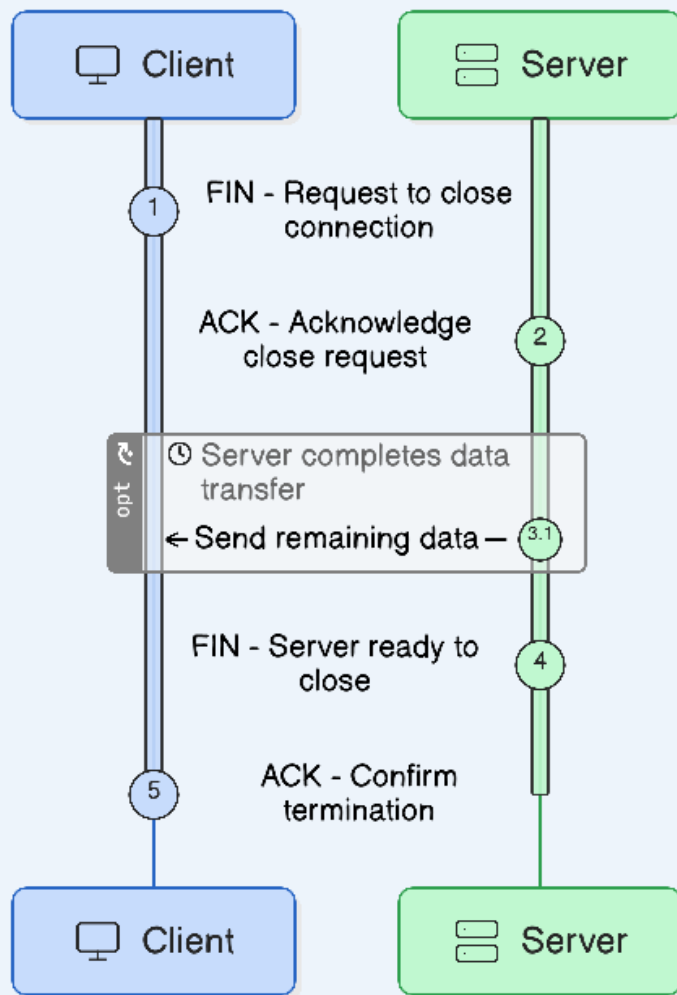
I. SYN (Synchronize): sender sends a SYN segment to connect

II. SYN-ACK (Synchronize-Acknowledge): receiver agrees to connect

III. ACK (Acknowledge): sender confirms connection established



TCP: TERMINATION

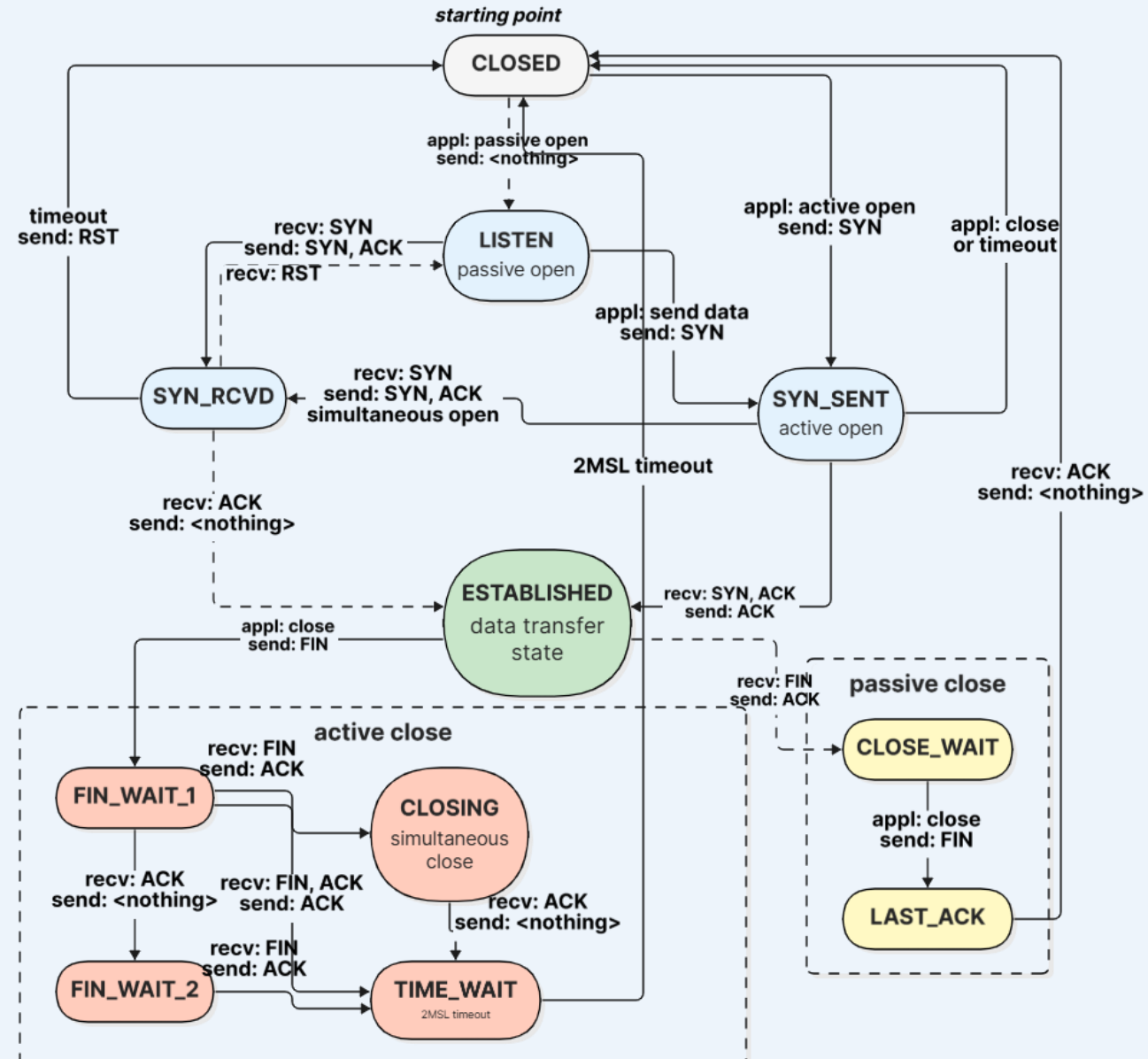


- I. **FIN (Finish):** The sender who wants to close the connection sends a FIN segment to the receiver.
- II. **ACK (Acknowledge):** The receiver acknowledges the FIN with an ACK.
- III. **FIN (Finish) from Receiver:** The receiver then sends its own FIN when it is ready to close the connection.
- IV. **ACK (Acknowledge):** The sender responds with an ACK, completing the termination.

TCP: STATE TRANSITION

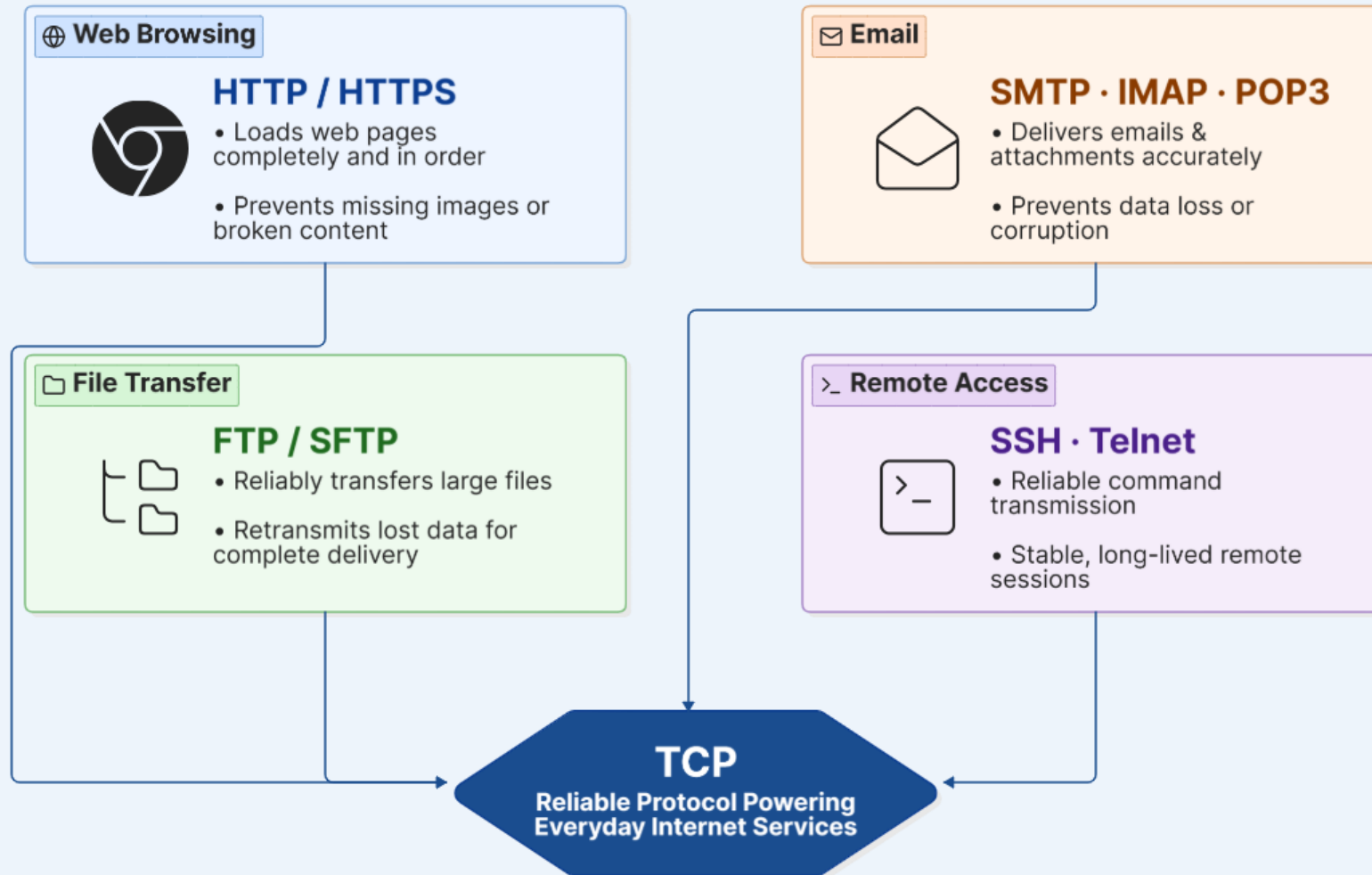
State	Description
CLOSED	There is no connection.
LISTEN	The server is waiting for calls from the client.
SYN-SENT	A connection request is sent; waiting for acknowledgment.
SYN-RCVD	A connection request is received.
ESTABLISHED	Connection is established.
FIN-WAIT-1	The application has requested the closing of the connection.
FIN-WAIT-2	The other side has accepted the closing of the connection.
TIME-WAIT	Waiting for retransmitted segments to die.
CLOSE-WAIT	The server is waiting for the application to close.
LAST-ACK	The server is waiting for the last acknowledgment.

TCP: STATE TRANSITION DIAGRAM



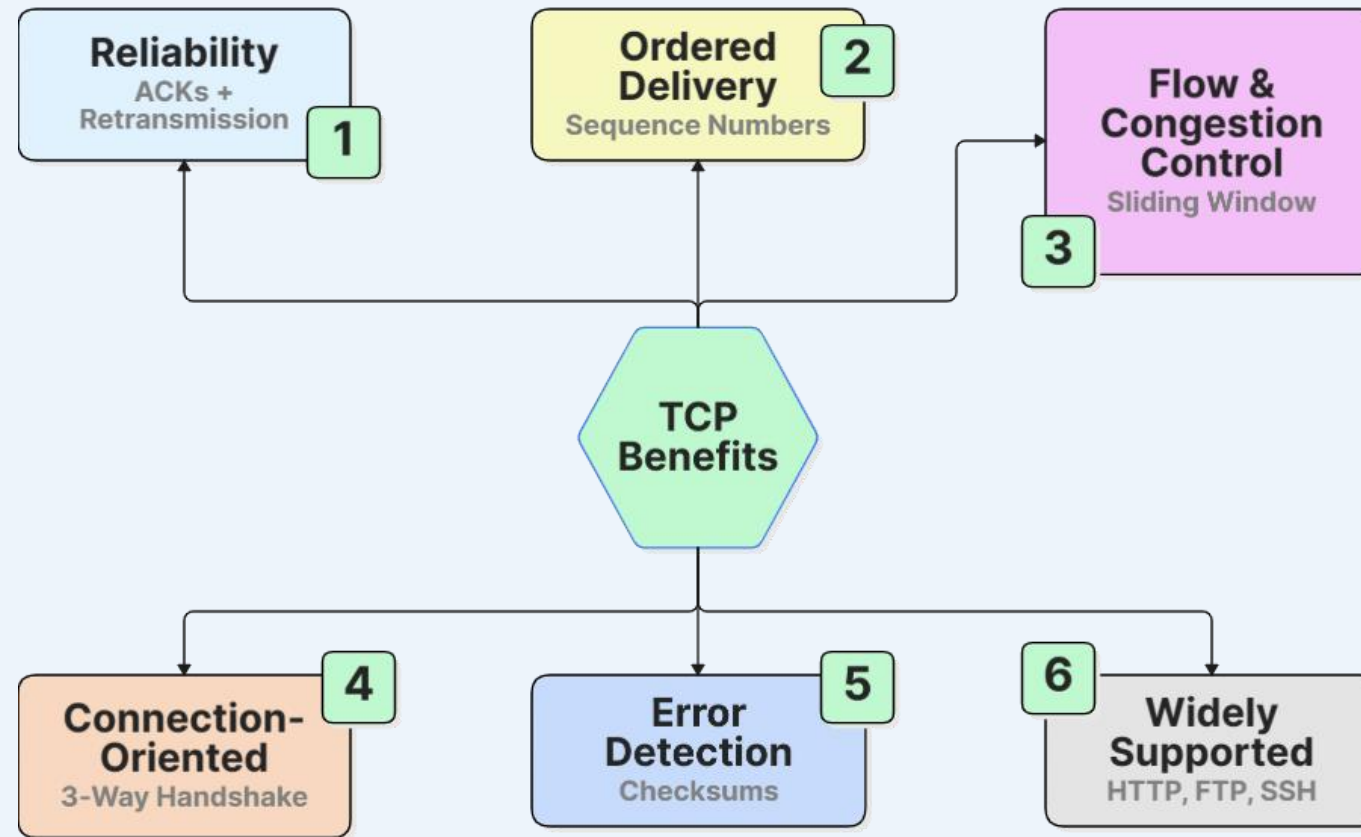


TCP APPLICATION





TCP ADVANTAGES





TCP LIMITATIONS

Overhead

Handshaking, ACKs & 20+ byte headers add overhead.

1

Slower Speed

Reliability mechanisms add latency vs. UDP.

2

Not Ideal for Real-Time

Retransmission delays hurt streaming, gaming & VoIP.

3

Connection Setup Time

Three-way handshake delays data transfer.

4

Resource Intensive

Buffers, timers & state consume memory and CPU.

5

Head-of-Line Blocking

Lost packet stalls all following packets.

6



REFERENCES

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